

- 3 Study this text title. What do you think it means?

Operating Systems: Hidden Software

Now read this text to check your answer and to find the answers to these questions:

- 1 What difference is there between applications software and operating systems?
- 2 Why is the supervisor program the most important operating system program?
- 3 What is the difference between resident and non-resident programs?
- 4 What are the main functions of an operating system?

When a brand new computer comes off the factory assembly line, it can do nothing. The hardware needs software to make it work. Are we talking about applications software such as wordprocessing or spreadsheet software? Partly. But an applications software package does not communicate directly with the hardware. Between the applications software and the hardware is a software interface – an operating system. An operating system is a set of programs that lies between applications software and the computer hardware.

The most important program in the operating system, the program that manages the operating system, is the supervisor program, most of which remains in memory and is thus referred to as resident. The supervisor controls the entire

operating system and loads into memory other operating system programs (called non-resident) from disk storage only as needed.

An operating system has three main functions: (1) manage the computer's resources, such as the central processing unit, memory, disk drives, and printers, (2) establish a user interface, and (3) execute and provide services for applications software. Keep in mind, however, that much of the work of an operating system is hidden from the user. In particular, the first listed function, managing the computer's resources, is taken care of without the user being aware of the details.

Furthermore, all input and output operations, although invoked by an applications program, are actually carried out by the operating system.

4 Complete the gaps in this summary of the text on operating systems using these linking words and phrases:

although *in addition*
because *such as*
but *therefore*

The user is aware of the effects of different applications programs operating systems are invisible to most users. They lie between applications programs, wordprocessing, and the hardware. The supervisor program is the most important. It remains in memory, it is referred to as resident. Others are called non-resident they are loaded into memory only when needed. Operating systems manage the computer's resources, the central processing unit., they establish a user interface, and execute and provide services for applications software. input and output operations are invoked by applications programs, they are carried out by the operating system.

LANGUAGE WORK

-ing form (1) as a noun; after prepositions

We can use the *-ing* form of the verb as a noun. It can be the subject, object, or complement of a sentence. For example:

- 1 *Managing* the computer's resources is an important function of the operating system.
- 2 The operating system starts *running* the user interface as soon as the PC is switched on.
- 3 Another function of the operating system is *executing* and *providing* services for applications software.

The *-ing* form is also used after prepositions. This includes *to* when it is a preposition and not part of the infinitive. For example:

- 4 *Without* the user *being* aware of the details, the operating system manages the computer's resources.
- 5 We begin *by focusing* on the interaction between a user and a PC operating system.
- 6 We look forward *to having* cheaper and faster computers.

5 Rewrite each of these sentences like this:

An important function of the operating system is to manage the computer's resources.

Managing the computer's resources is an important function of the operating system.

- 1 One task of the supervisor program is to load into memory non-resident programs as required.
- 2 The role of the operating system is to communicate directly with the hardware.
- 3 One of the key functions of the operating system is to establish a user interface.
- 4 An additional role is to provide services for applications software.
- 5 Part of the work of mainframe operating systems is to support multiple programs and users.
- 6 The task in most cases is to facilitate interaction between a single user and a PC.
- 7 One of the most important functions of a computer is to process large amounts of data quickly.
- 8 The main reason for installing more memory is to allow the computer to process data faster.

6 Complete these sentences with the correct form of the verb: infinitive or *-ing* form.

- 1 Don't switch off without (close down) your PC.
- 2 I want to (upgrade) my computer.
- 3 He can't get used to (log on) with a password.
- 4 You can find information on the Internet by (use) a search engine.
- 5 He objected to (pay) expensive telephone calls for Internet access.
- 6 He tried to (hack into) the system without (know) the password.
- 7 You needn't learn how to (program) in HTML before (design) webpages.
- 8 I look forward to (input) data by voice instead of (use) a keyboard.